Printed Pages: 01 **Sub Code: BOP-231**

Paper Id: 233038 **Roll No.** | | | | | | | | | | |

B.PHARM.
(SEM III) THEORY EXAMINATION 2022-23
PHARMACEUTICAL CHEMISTRY-III (HETEROCYCLIC & BIOORGANIC CHEMISTRY)

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 7 = 14

(a) Discuss the structure and uses of Furan.
(b) Discuss the chemistry of Pyridine
(c) What is non-reducing sugar?
(d) Differentiate between Glucose and Fructose.
(e) Define amino acids.
(f) Outline the physiological functions of Thiamine.
(g) Define polymerization.

SECTION B

2. Attempt any *three* of the following: 7 x3= 21

(a) Outline the nomenclature of heterocyclic compounds with suitable examples.
(b) Define carbohydrates. Classify them with suitable examples.
(c) Illustrate the types of protein structure.
(d) Write a detailed note on genetic code.
(e) Give the structure elucidation and physiological functions of niacin.

SECTION C

3. Attempt any *one* part of the following: 7 x1= 7

(a) Compare the basicity of Pyrrole and Pyrazole.
(b) Explain the chemistry, method of preparation and pharmaceutical applications of Imidazole.

4. Attempt any *one* part of the following: 7 x 1 = 7

(a) Give the structure elucidation of Glucose in detail.
(b) Define polysaccharides. Classify them and write their importance.

5. Attempt any *one* part of the following: 7 x 1 = 7

(a) Outline solid phase peptide synthesis.
(b) Discuss about protein separation and purification techniques.

6. Attempt any *one* part of the following: 7 x 1 = 7

(a) Classify nucleic acid with suitable example. Outline the functions of DNA and RNA.
(b) Illustrate about the acid value and saponification value in details.

7. Attempt any *one* part of the following: 7 x1= 7

(a) Define Vitamins. Outline the classification of Vitamins.
(b) Write a detailed note on polymers.

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